
Where Alignment Surprises: Predicting Per-Token Base $\leftrightarrow$ Instruct Divergence and Reading Its Residuals

Ari Holtzman and NeuriCo

Abstract

Instruction tuning rewrites a base model’s next-token distribution, but most of that change is anticipatable. We ask a sharper question than prior work: where does alignment change the distribution in ways that a learned predictor cannot anticipate from simple base-side context, and what kinds of tokens live there? The Superficial Alignment Hypothesis holds that alignment mostly adjusts surface style, but no one has trained a predictor of per-token divergence and studied the errors it makes. We do exactly that. For each instruction we greedily decode an instruct response, teacher-force the identical sequence through both base and instruct models under the same chat-formatted context to isolate the weight difference, and at every response position compute the target divergence plus cheap base-side features (position, base entropy, base top-1 probability, emitted-token log-probability, token class). We fit a gradient-boosted predictor on one corpus and read its residuals on a disjoint one, across three Qwen2.5 scales (0.5B, 1.5B, 7B). The predictor explains roughly 60–68% of per-token divergence cross-corpus, driven almost entirely by base-side uncertainty; position alone is nearly useless. The residual, where the model is provably wrong, is not noise: it concentrates significantly on safety prompts and on discourse and structural pivots, persona and identity tokens (the model naming its maker, framing a knowledge cutoff), and refusal-recovery moves. This refines the Superficial Alignment Hypothesis: the average shift is predictable and largely superficial, but the unpredictable residual is substantive, alignment-specific, and localizable with no labels.

1 Introduction

Instruction tuning turns a next-token predictor into an assistant, yet most of what it changes is cosmetic. When we line up a base model and its instruction-tuned sibling on the same chat-formatted text, their top-1 predictions agree on 83–85% of response tokens, and the mean per-token $KL(P_{\text{inst}} \parallel P_{\text{base}})$ between them is only 0.21–0.24 nats. The Superficial Alignment Hypothesis [Zhou et al., 2023] and analyses of the post-tuning distribution shift [Lin et al., 2023] argue that alignment mostly reweights a thin band of stylistic and formatting tokens that the base model already knew how to produce. If that were the whole story, the gap between base and instruct would be almost entirely a matter of surface manners.

But recent work pushes back: instruction tuning also reshapes substantive behavior, and the “superficial” framing undersells what changes [Wu et al., 2024]. This leaves a sharp empirical question that prior work has not answered at the resolution of individual tokens. **Where does alignment do something a simple model of the base could not have guessed?** If a divergence is predictable from cheap base-side context—this token is low-confidence, that one is high-entropy—then it is the boring, stylistic part of alignment doing its job. The interesting part is the divergence that remains *unpredictable*: the positions where the instruct model moves probability mass in a way the base context gives no warning of. We argue that is precisely where alignment surprises, and that reading those surprises tells us what instruction tuning actually installs.

Our approach. We make this precise by turning “surprise” into a residual. For each instruction we greedily decode an ≤ 80 -token response with the instruct model, then teacher-force the identical token sequence (chat prompt plus response) through *both* the base and the instruct model under the same chat-formatted context. This isolates the weight difference alone: same tokens, same context, only the parameters differ. At each response position we record the target $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ over the full vocabulary together with a handful of base-side features—position, base entropy, base top-1 probability, the base log-probability of the emitted token, token-class flags, and token length. We train a gradient-boosted tree regressor (GBT) to predict $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ from those features on DOLLY-15K [Conover et al., 2023], then study the *residuals*—actual minus predicted $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ —on the disjoint JUST-EVAL-INSTRUCT set [Lin et al., 2023], 1000 instructions of which 200 carry safety tags. We repeat the whole pipeline across three scales of the QWEN2.5 family [Team, 2025]: QWEN2.5-0.5B, QWEN2.5-1.5B, and QWEN2.5-7B, base and instruct.

What we find. Per-token divergence is largely predictable. The GBT reaches cross-corpus $R^2 \approx 0.60\text{--}0.68$ (0.599 / 0.680 / 0.626 across the three pairs), with Spearman ρ of 0.82–0.89 between predicted and actual $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$. Two features carry almost all the signal—base token log-probability and base entropy—while a position-only baseline is essentially useless ($R^2 \leq 0.05$) and every stylistic or lexical token flag contributes under 0.03 importance. So most of the base→instruct shift really is anticipatable from how confident the base model already was. The residual, however, is not noise: it is structured and alignment-specific. Safety-tagged tokens carry a significantly positive mean residual in all three pairs (e.g., QWEN2.5-0.5B: +0.138 vs. −0.012 for regular tokens), and the largest residuals concentrate on safety language, discourse and structural pivots, and persona/identity tokens—not the generic stylistic tokens that dominate the predictable shift. A telling case: after a refusal, the instruct model emits “If you have any other questions...”, but the base ranks “If” only third, yielding $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}}) \approx 19.9$ —a refusal-recovery pivot the base context never signals. Identity tokens like “Alibaba” (Qwen’s maker) and knowledge-cutoff framing like “October” surface the same way.

This reframes the superficial-alignment debate quantitatively: alignment is *mostly* predictable reweighting, but the unpredictable remainder is exactly the safety, framing, and persona behavior that motivates alignment in the first place.

Contributions.

- We operationalize per-token base↔instruct divergence as a teacher-forced $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ under a shared chat context, isolating the weight difference from any decoding or prompting confound.
- We train a base-side GBT predictor of this divergence and show it captures most of the shift (cross-corpus $R^2 \approx 0.60\text{--}0.68$, Spearman ρ up to 0.89), driven almost entirely by base confidence and entropy rather than position or lexical flags.
- We define and characterize the *residual*—the unpredictable divergence—and demonstrate it is structured rather than noise (Kruskal–Wallis across token classes $p < 10^{-22}$ in every pair).
- We show across three model scales that the residual is alignment-specific: it concentrates on safety (significant in all 3 pairs), discourse/structural pivots, and persona/identity tokens, with the safety effect shrinking as scale grows.

We review related work on the alignment delta and distribution shift in section 2, detail our teacher-forcing and prediction setup in section 3, report predictability and residual analyses in section 4, and discuss implications in section 5.

2 Related Work

The Superficial Alignment Hypothesis. A growing line of work asks how much instruction tuning really changes a base model. Zhou et al. [2023] articulate the Superficial Alignment Hypothesis: a model’s knowledge and capabilities are learned almost entirely during pretraining, and alignment mainly teaches it which subdistribution of formats and styles to surface. This view is supported by the strong performance of LIMA from a thousand curated examples, and by in-context alignment results suggesting that a few demonstrations recover much of the aligned behavior [Zhao et al., 2024, Lin et al., 2023]. Other work pushes back: Wu et al. [2024] report cases where fine-tuning injects new behavior that few-shot prompting cannot reproduce, arguing the shift is not purely superficial. We do not take a side in this debate at the level of aggregate behavior. Instead, we operationalize

“superficial” *per token*: if alignment were entirely a stylistic re-weighting, the $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ at each position would be predictable from cheap base-side context. Our GBT predictor measures exactly how much of the shift is this superficial component, and our *residual* isolates the part that is not.

Token-distribution-shift analysis. Closest to us is URIAL [Lin et al., 2023], which compares aligned and base models token by token. They introduce a base-rank diagnostic, ranking each token the aligned model emits within the base model’s distribution, and find that the two models already agree on the top-1 token roughly 78% of the time, that disagreement concentrates in a small set of stylistic and discourse markers (transition words, safety disclaimers, formatting), and that the shift decays sharply as a response proceeds. We reproduce all three findings at the distributional level: top-1 agreement of 0.83–0.85, mean per-token $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ of 0.21–0.24 nats, and KL that decays with position (on QWEN2.5-0.5B, mean KL falls from 0.68 at positions 0–4 to 0.12 past position 40). Lin et al. [2023] read off these shifted tokens by eye to argue alignment is superficial. Unlike URIAL, we do not stop at describing *which* tokens shift on average; we fit an explicit predictor of the shift and ask what is *left over* once the predictable, stylistic component is removed.

Logit-difference behavior signals. A parallel thread treats the difference between aligned and base *logits* as a usable signal rather than a nuisance. Emulated Fine-Tuning [Mitchell et al., 2023] formalizes the behavior $\delta \log \pi_{\text{inst}} - \log \pi_{\text{base}}$ and shows it can be transplanted across model scales; proxy-tuning [Liu et al., 2024] adds a small tuned model’s logit difference to a large base model to steer it without training the large model. Contrastive-decoding methods exploit the same gap for generation quality and reasoning: Li et al. [2022] and O’Brien and Lewis [2023] contrast a strong and a weak model, and DoLa [Chuang et al., 2023] contrasts early and late layers of a single model. These methods *use* the logit difference to decode; they do not ask which parts of it are anticipatable. Unlike them, we treat the per-token divergence as a *target to predict* from base-side context, so that the prediction error, not the difference itself, becomes the object of study.

Probability concentration under alignment. Yang et al. [2026] study alignment through the branching factor and argue that instruction tuning sharpens the next-token distribution, concentrating mass and surfacing low-entropy continuations that already existed latently in the base model. This gives a mechanistic reason why so much of the shift should be predictable from base-side uncertainty: where the base model is already confident, alignment has little room to move, and where it is uncertain, alignment often collapses onto a preferred continuation. Our feature-importance results bear this out, with base top-1 log-probability and base entropy dominating the predictor. But sharpening is a statement about the *average* or *typical* token; it does not say where alignment overrides the base model’s preference in ways uncertainty cannot forecast. That override is precisely what our residual captures.

Transferable weight deltas and interpretability. Finally, recent work shows the alignment weight delta is itself a portable object: Wu et al. [2025] transfer a fine-tuning delta between models, treating it as a reusable patch. Interpretability tools such as LogitLens4LLMs [Wang et al., 2025] project internal states to the vocabulary to read where and how a model commits to a token. We work in the same spirit of reading the alignment difference directly in probability space, but at the granularity of individual response positions, and we use teacher forcing of the *identical* chat-formatted sequence through both models so that the only thing that varies is the weights.

Positioning. Across all of this work, the alignment shift is characterized by its *average* or its *predictable* structure: which tokens move, how much mass concentrates, how the delta transfers. None of it trains an explicit predictor of the per-token divergence and then studies the *residual* of that predictor. That residual, the part of $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ that a learned model genuinely cannot anticipate from simple base-side context, is our object of study, and we show it is neither noise nor the generic stylistic tokens everyone has already cataloged, but alignment-specific structure: safety refusals, discourse and structural pivots, and persona and identity injections.

3 Methodology

Our goal is to localize, token by token, where an instruct model’s next-token distribution departs from its base model, and to ask how much of that departure a learned predictor can anticipate from cheap base-side context. We first define the per-token divergence target (section 3), then the models

and corpora, the divergence predictor and its baselines, the residual analysis protocol, and finally the details needed to reproduce every number.

3.1 Per-token divergence

Shared-context teacher forcing. For each instruction we first format it with the model’s chat template and greedily decode a response of up to 80 new tokens from the *instruct* model. Let $o_{1:T}$ denote that response and let c denote the chat-formatted prompt. We then teacher-force the *identical* full sequence $(c, o_{1:T})$ through both the base and the instruct model. Crucially, both models see the same chat-formatted context c : we do not strip the chat template for the base model. This matters. Differences in formatting would otherwise confound the comparison; by holding the context fixed and identical, the only thing that varies between the two forward passes is the network weights. The per-position distributions we compare therefore isolate the *weight delta* induced by instruction tuning, in the same spirit as the behavior delta $\log \pi_{\text{inst}} - \log \pi_{\text{base}}$ studied by Mitchell et al. [2023] and the logit difference exploited by Liu et al. [2024].

KL divergence. At each response position $t \in \{1, \dots, T\}$ both models induce a next-token distribution over the full vocabulary V given the shared prefix $(c, o_{<t})$. We write $P_{\text{inst}}(\cdot \mid c, o_{<t})$ and $P_{\text{base}}(\cdot \mid c, o_{<t})$ for these distributions and define our primary divergence target as the forward KL,

$$\text{KL}_t = \text{KL}(P_{\text{inst}} \parallel P_{\text{base}}) = \sum_{v \in V} P_{\text{inst}}(v \mid c, o_{<t}) \log \frac{P_{\text{inst}}(v \mid c, o_{<t})}{P_{\text{base}}(v \mid c, o_{<t})}, \quad (1)$$

computed over the entire vocabulary (no top- k truncation) in nats. We use the asymmetric forward direction deliberately: it weights mass where the *instruct* model is confident, so KL_t is large exactly at the tokens the aligned model wants to emit but the base model would not.

Cross-checks. KL is a distributional summary; to interpret what it captures we also record two scalar cross-checks at each position. The first is the base-side rank η_t of the emitted token o_t under P_{base} ($\eta_t = 1$ when the base model would itself have greedily emitted o_t). The second is the per-token log-probability delta $\log P_{\text{inst}}(o_t \mid c, o_{<t}) - \log P_{\text{base}}(o_t \mid c, o_{<t})$, the pointwise version of the EFT/proxy-tuning behavior delta. Where KL_t is large we expect η_t to climb and the log-probability delta to swing positive; these give an interpretable handle on individual high-divergence tokens.

3.2 Models and data

Model pairs. We study three QWEN2.5 base/instruct pairs spanning more than an order of magnitude in scale: QWEN2.5-0.5B, QWEN2.5-1.5B, and QWEN2.5-7B, each as its released base and instruction-tuned checkpoint [Team, 2025]. Within each pair the base and instruct checkpoints share architecture, tokenizer, and vocabulary, so equation 1 is well defined and the comparison reduces cleanly to the weight difference. Spanning 0.5B to 7B lets us ask whether the structure we find is an artifact of small models or persists with scale.

Corpora. We train the divergence predictor and analyze its residuals on *disjoint* instruction sets, so that no instruction seen in fitting reappears in analysis. The predictor is fit on instructions from DOLLY-15K [Conover et al., 2023]. All residual analysis is performed on JUST-EVAL-INSTRUCT [Lin et al., 2023], the 1000-instruction evaluation set introduced with URIAL, of which 200 instructions carry a safety tag. The strict train/analyze split across two independently constructed corpora rules out leakage and means the structure we report in the residuals is not memorized from the fitting data. For each model pair we collect roughly 1200–1500 instruction–response trajectories, each contributing up to 80 per-token examples.

3.3 Divergence predictor and baselines

Base-side features. The predictor is restricted to features computable from the *base* model and the surface form of the emitted token alone—it never sees P_{inst} . For each position t we extract: the response position t ; the base entropy $H(P_{\text{base}}(\cdot \mid c, o_{<t}))$; the base top-1 probability $\max_v P_{\text{base}}(v \mid c, o_{<t})$; the base log-probability of the emitted token $\log P_{\text{base}}(o_t \mid c, o_{<t})$; a set of token-class flags (e.g., punctuation, whitespace/formatting, digits, alphabetic word pieces); and the token string length in characters. This feature set deliberately excludes anything that would require running the instruct

model, so a low residual would mean alignment’s effect at that token is anticipable from how the base model already behaves there.

Predictor. Our divergence predictor, GBT, is a `HistGradientBoostingRegressor` [Pedregosa et al., 2011]—a histogram-based gradient-boosted tree ensemble in the `LightGBM` lineage [Ke et al., 2017]—trained to regress KL_t on the base-side features above. We run up to 600 boosting iterations with early stopping on a held-out validation split. We fit one GBT per model pair on DOLLY-15K and report cross-corpus R^2 and Spearman ρ between predicted and actual KL on JUST-EVAL-INSTRUCT.

Baselines. To establish how much each individual cue explains on its own, we compare GBT against three single-feature linear regressions: a *position-only* baseline (KL from t alone), an *entropy-only* baseline (KL from base entropy), and a *base-logprob-only* baseline (KL from $\log P_{\text{base}}(o_t)$). These isolate the contribution of position, base uncertainty, and base surprise respectively, and bracket the gap that the full nonlinear predictor closes.

3.4 Residual protocol

Definition. For every response token in JUST-EVAL-INSTRUCT we compute the residual,

$$r_t = KL_t - \widehat{KL}_t,$$

the actual minus the GBT-predicted divergence, using the predictor fit on DOLLY-15K. A positive r_t marks a token where instruction tuning shifted the distribution *more* than the base-side context anticipates: an alignment surprise. Because the predictor already absorbs the generic, anticipable component of the shift, the residual is where we look for structure.

Slicing. We slice the residuals three ways: by instruction category (safety-tagged vs. regular), by token class, and by position bin. For each slice we ask whether residuals are centered away from zero and whether they differ across slices.

Statistics. We test residual structure with nonparametric tests throughout, since residuals are heavy-tailed. We run a Kruskal–Wallis omnibus test across token classes to ask whether any class carries systematically different residuals. For the safety contrast we run a one-sided Mann–Whitney U test of the hypothesis that safety-tagged residuals exceed regular ones, and report the rank-biserial correlation as a scale-free effect size alongside the mean residual per slice. We use $\alpha = 0.05$ throughout; given the sample sizes we also report exact p -values.

3.5 Reproducibility

All decoding is greedy with a fixed seed of 42, so trajectories are deterministic. Forward passes use `fp16`; for each pair we place the base model on `cuda:0` and the instruct model on `cuda:1` to compute both distributions for a position without reloading weights. Experiments run on $4\times$ RTX A6000 (48 GB) GPUs. End-to-end wall-clock is roughly 35 minutes for the QWEN2.5-0.5B and QWEN2.5-1.5B pairs together and a further 15 minutes for QWEN2.5-7B. The software stack is Python 3.12, torch 2.12, transformers 5.10.2, scikit-learn 1.9, and datasets 5.0.

4 Results

We organize the evidence around the three hypotheses, after first confirming that our measurement pipeline reproduces known facts about the base-to-instruct shift. All numbers are computed on JUST-EVAL-INSTRUCT response tokens, with the predictor trained on the disjoint DOLLY-15K corpus.

4.1 Measurement validation

We reproduce URIAL’s stylized facts. Before reading residuals, we check that teacher-forcing both models under the identical chat-formatted context recovers the qualitative picture established by Lin et al. [2023]. Across the three QWEN2.5 pairs, the base and instruct models agree on the top-1 next token at **83–85%** of response positions, closely matching URIAL’s $\sim 78\%$ (their estimate is on a different family and decoding setup, so near-agreement is the expected outcome). The mean per-token $KL(P_{\text{inst}} \parallel P_{\text{base}})$ is small, **0.21–0.24 nats**, confirming that most of the response is generated

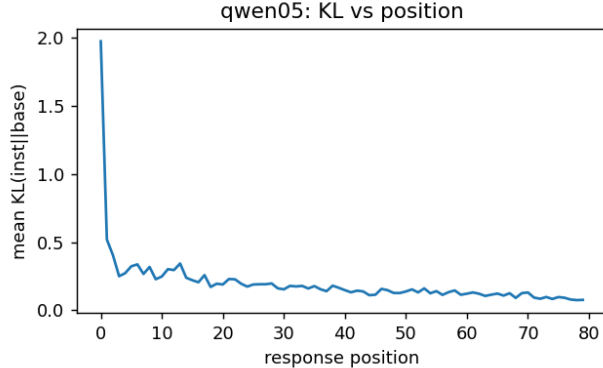


Figure 1: Mean per-token $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ as a function of response position for QWEN2.5-0.5B. Divergence is strongly front-loaded, decaying from ~ 0.68 nats in the first few tokens to ~ 0.12 nats after position 40. Instruction tuning chiefly rewrites the start of a response. Shaded band shows the standard error within each position bin.

Pair	top-1 agree	mean KL	Baseline R^2 (val)			GBT R^2		Spearman ρ
			pos-only	entropy-only	base-logp-only	val	x-corpus	
QWEN2.5-0.5B	0.829	0.207	0.015	0.196	0.398	0.704	0.599	0.821
QWEN2.5-1.5B	0.852	0.214	0.046	0.394	0.382	0.693	0.680	0.849
QWEN2.5-7B	0.826	0.240	0.022	0.281	0.510	0.621	0.626	0.885

Table 1: **Per-token KL is largely predictable from base-side context, and position alone is nearly useless.** Top-1 agreement and mean per-token $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ (nats) measured on JUST-EVAL-INSTRUCT. Predictor R^2 on a held-out validation split of DOLLY-15K (val) and on the disjoint JUST-EVAL-INSTRUCT corpus (x-corpus). The gradient-boosted predictor (GBT) explains ~ 60 – 68% of cross-corpus KL variance, far above every single-feature baseline; position-only regression is near-zero ($R^2 \leq 0.05$). Spearman ρ is between predicted and actual KL on JUST-EVAL-INSTRUCT. Best per-row GBT values in **bold**.

by tokens on which the two models barely disagree. This is the backdrop against which a learned predictor has any hope of working: the shift is concentrated, not uniform.

The shift is front-loaded. Figure 1 shows that divergence is sharply position-dependent. For QWEN2.5-0.5B, mean $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ falls from **0.68 nats at positions 0–4** to **0.12 nats at positions 40+**, a roughly $6\times$ decay. Instruction tuning rewrites the opening of a response—where the model commits to a stance, a format, or a refusal—and then largely defers to the base distribution once the response is underway. **Is position therefore the story?** We show next that it is not: position alone explains almost none of the variance once we condition on what the base model was already doing.

4.2 H1: per-token KL is largely predictable

A simple base-side predictor captures most of the divergence. We train the GBT on DOLLY-15K and evaluate on JUST-EVAL-INSTRUCT, so any signal it captures is cross-corpus and cannot be leakage. Table 1 reports the result: cross-corpus R^2 is **0.599** (QWEN2.5-0.5B), **0.680** (QWEN2.5-1.5B), and **0.626** (QWEN2.5-7B), with validation R^2 in the same 0.62–0.70 range. Rank agreement is even stronger—Spearman $\rho(\text{pred}, \text{actual})$ reaches **0.82 / 0.85 / 0.89** as we scale. Two-thirds of the per-token variance in $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ is anticipated from features that never look at the instruct model.

The signal is base entropy and base surprise, not position or token type. Figure 2 and Table 2 make the source of the predictability explicit. The base model’s log-probability of the emitted token (`base_tok_logprob`) and its predictive entropy (`base_entropy`) dominate, with importances of **~ 0.4 – 0.6 each**; position contributes only ~ 0.03 – 0.08 , and every stylistic, lexical, and token-class flag sits below 0.03. Single-feature baselines confirm the ranking: a position-only model is near

Feature	QWEN2.5-0.5B	QWEN2.5-1.5B	QWEN2.5-7B
base_tok_logprob	0.603	0.399	0.609
base_entropy	0.496	0.657	0.139
base_top1_prob	0.273	0.139	0.427
pos	0.076	0.077	0.033
tok_len	0.012	0.016	0.017
is_punct	0.011	0.002	0.026
is_stylistic	0.010	0.015	0.003
is_wordlead	0.004	0.015	0.006
is_number	0.001	0.000	0.001

Table 2: **Predictive structure is base-model uncertainty, not position or lexical class.** Gradient-boosting feature importance (gain) for the GBT divergence predictor across all three pairs. Base token surprise (base_tok_logprob) and base entropy (base_entropy) dominate; response position (pos) and every stylistic/lexical flag contribute little. This refines URIAL’s “KL decays with position”: the decay is mostly *mediated by entropy decay*, not position per se. Largest per-column value in **bold**.

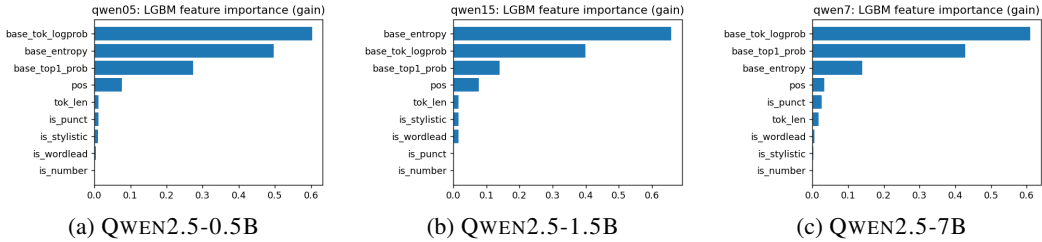


Figure 2: GBT feature importances for predicting per-token $\text{KL}(P_{\text{inst}} \| P_{\text{base}})$, across all three model pairs. Two base-side quantities—the base log-probability of the emitted token and the base predictive entropy—dominate, while position and all stylistic, lexical, and token-class flags are negligible (<0.03). The predictable component of the base-to-instruct shift is governed by where the base model is already uncertain.

useless (R^2 **0.015–0.046**), an entropy-only model reaches 0.196/0.394/0.281, and a base-logprob-only model reaches 0.398/0.382/0.510. **What does this refine?** URIAL frames the shift as decaying with position; we find that decay is *entropy-mediated*. Early tokens diverge because the base model is uncertain there; once the base distribution sharpens, instruction tuning has little room to move it. Position is a proxy for surprise, not a cause.

The predictor is well-calibrated. Figure 3 shows predicted versus actual $\text{KL}(P_{\text{inst}} \| P_{\text{base}})$ for QWEN2.5-0.5B; points track the diagonal across the bulk of the range, so the residuals we analyze next are genuine prediction errors rather than systematic miscalibration. This is what licenses reading residuals as “surprises.”

4.3 H2: the residual is structured

The leftover variance is not noise. If the residual were homogeneous prediction error, it would be unrelated to token identity. It is not. A Kruskal–Wallis test across token classes rejects equality of residual distributions at $p < 10^{-22}$ **for every pair** (QWEN2.5-0.5B 2.9×10^{-29} , QWEN2.5-1.5B 1.2×10^{-23} , QWEN2.5-7B 1.3×10^{-88}). The residual also varies systematically across position bins even though position is itself an input feature the GBT could already exploit—meaning the structure that remains is not capturable by position alone. The model has learned the smooth, entropy-driven backbone of the shift and left behind something organized that those features cannot reach.

4.4 H3: the residual is alignment-specific

Safety prompts are systematically under-predicted. The structure in the residual is exactly the content one would attribute to alignment, not to generic style. Table 3 compares the 200 safety-tagged JUST-EVAL-INSTRUCT instructions against the rest. Safety tokens carry a positive

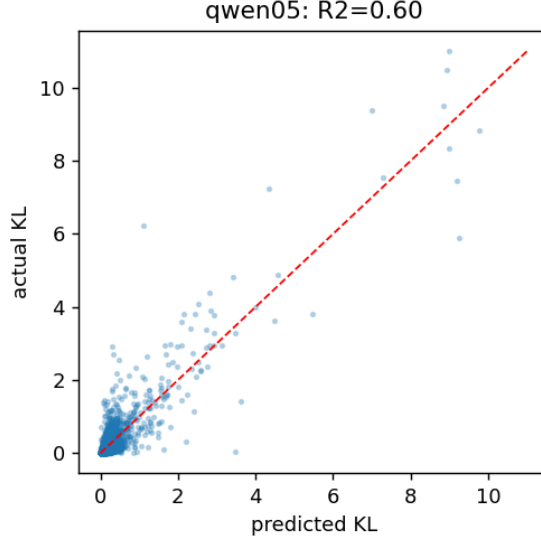


Figure 3: Predicted versus actual per-token $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ on held-out JUST-EVAL-INSTRUCT tokens for QWEN2.5-0.5B. Predictions track the diagonal across the bulk of the range, confirming the GBT is calibrated rather than merely correlated, so the residual (actual – predicted) is a clean signal of what the base-side features fail to anticipate.

Pair	mean resid (safety)	mean resid (regular)	rank-biserial	p
QWEN2.5-0.5B	+0.138	−0.012	0.322	≈ 0
QWEN2.5-1.5B	+0.085	−0.015	0.302	3×10^{-284}
QWEN2.5-7B	+0.061	−0.012	0.141	2×10^{-164}

Table 3: **The residual concentrates on safety prompts in every pair.** Mean prediction residual (actual KL – predicted KL) on safety vs. regular JUST-EVAL-INSTRUCT prompts, with one-sided Mann–Whitney p and rank-biserial effect size. Safety prompts are exactly where the predictor most *under*-estimates divergence. The effect shrinks with scale ($0.322 \rightarrow 0.302 \rightarrow 0.141$) but remains overwhelmingly significant. Larger safety mean residual in **bold**.

mean residual while regular tokens carry a near-zero or slightly negative one: **+0.138 vs −0.012** (QWEN2.5-0.5B, rank-biserial 0.322), **+0.085 vs −0.015** (QWEN2.5-1.5B, 0.302), and **+0.061 vs −0.012** (QWEN2.5-7B, 0.141). One-sided Mann–Whitney tests are overwhelming ($p \approx 0$ for QWEN2.5-0.5B, 3×10^{-284} for QWEN2.5-1.5B, 2×10^{-164} for QWEN2.5-7B). The predictor consistently *underestimates* divergence on safety content: the instruct model moves more than its base-side surprise would imply. The effect shrinks with scale—larger base models already place more mass on safety-consistent continuations—but remains highly significant everywhere.

Two interpretable token families drive the residual. Table 4 lists the highest-residual token strings. They fall into two recognizable groups. The first is *discourse and structural pivots*: tokens like *Additionally*, *However*, *Instead*, *Specifically*, *Such*, and—most tellingly—refusal-recovery openers. In a concrete QWEN2.5-0.5B case, after declining a request the instruct model emits “If you have any other questions...”; the base model ranks *If* only third at that position, and the per-token $\text{KL}(P_{\text{inst}} \parallel P_{\text{base}})$ spikes to ~ 19.9 nats. This is a learned move—softening a refusal by re-offering help—that the base distribution gives no hint of. The second group is *safety, persona, identity, and helpfulness injections*: *Safety*, *guidelines*, *violence*, *politics*, *discuss*, *provide*, *recommend*, alongside *assistant* and *Alibaba* (Qwen’s maker, an identity injection) and *October* (knowledge-cutoff framing). Figure 4 aggregates these into per-category residual distributions, and Figure 5 shows the picture is consistent across all three pairs: high predictability of the bulk shift, a structured residual, and that residual concentrated on alignment-specific tokens. Crucially, the stylistic tokens that dominate the *predictable* shift (punctuation, function words, formatting) do *not* appear here—the surprise is alignment, not style.

Pair	Family	Highest-residual token strings (mean residual, $n \geq 5$)
QWEN2.5-0.5B	Discourse pivots Safety / sensitive	Additionally, If, However, The, Building, Subject politics, sex, violence, encourage, distress, sensitive, political
QWEN2.5-7B	Discourse pivots Safety / persona / helpfulness	If, Such, However, Instead, Specifically, For Safety, guidelines, assistant, Alibaba, October, provide, recommend, discuss

Table 4: **The unpredictable residual recovers exactly what alignment installs.** Token strings with the highest mean residual on JUST-EVAL-INSTRUCT, grouped into two interpretable families: (i) discourse/structural pivots at clause boundaries (e.g., refusal-recovery openers such as If), and (ii) safety/persona/identity/helpfulness injections (e.g., Safety, guidelines, Alibaba — Qwen’s maker, an identity injection). Neither family is the generic stylistic token (Here, However) that the predictor already accounts for.

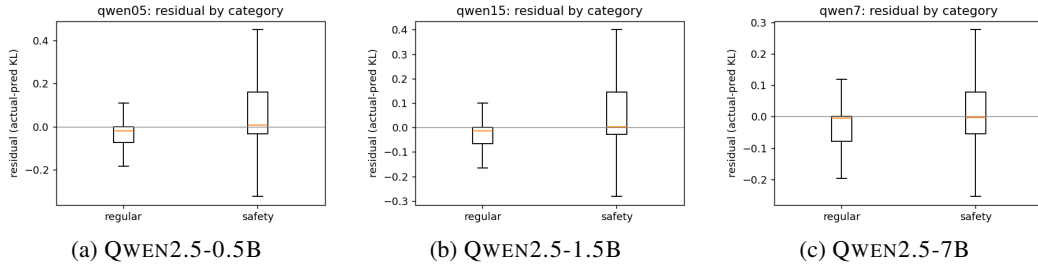


Figure 4: Mean residual (actual — predicted $\text{KL}(P_{\text{inst}} \| P_{\text{base}})$) by token category, across all three model pairs. Safety, discourse-pivot, and persona/identity categories sit well above zero, while generic stylistic and lexical categories sit at or below it. The base-side predictor systematically under-predicts exactly the alignment-relevant tokens.

5 Discussion

Where the predictor is wrong is where the interesting alignment lives. Our central finding is a useful inversion. A small GBT trained only on base-side features predicts per-token $\text{KL}(P_{\text{inst}} \| P_{\text{base}})$ well (section 4, $R^2 \approx 0.60\text{--}0.68$), so most of the base $\leftrightarrow$ instruct shift is anticipatable from how uncertain the base model already was. The payoff is not the fit but the miss. When we read the residual—actual minus predicted KL—on the disjoint JUST-EVAL-INSTRUCT corpus, the tokens that surprise the predictor are not random and not generic. They are refusal-recovery pivots (the instruct model emitting “If you have any other questions...” where the base model ranks “If” only third, $\text{KL} \approx 19.9$), persona and identity injections (“Alibaba,” Qwen’s maker), knowledge-cutoff framing (“October”), and explicit safety vocabulary (“violence,” “politics,” “guidelines,” “Instead,” “recommend”). We surfaced these with no safety labels, no persona annotations, and no discourse parser. The residual itself does the sorting. This is the practical content of H2 and H3: the unexplained part of alignment is structured, and that structure is alignment-specific rather than stylistic.

A quantitative refinement of the Superficial Alignment Hypothesis. Zhou et al. [2023] argued that alignment mostly teaches format and style already latent in the base model. Our results let us be precise about what “mostly” means and where it breaks. The bulk of $\text{KL}(P_{\text{inst}} \| P_{\text{base}})$ is explained by base uncertainty—base_tok_logprob and base_entropy carry nearly all of the predictor’s importance (table 2), while every stylistic and lexical flag sits below 0.03. So the part of the distribution shift that is genuinely superficial is exactly the part a base-side predictor can already see: where the base model was unsure, instruction tuning resolves the choice in a stylistically predictable direction. This is consistent with Yang et al. [2026], whose branching-factor view holds that alignment sharpens an already-uncertain distribution rather than rewriting it. But the residual is the non-superficial remainder, and it is not negligible—it concentrates on safety, identity, and discourse control. This is the regime Wu et al. [2024] point to as counter-evidence: behaviors that base-side context cannot reconstruct. We do not have to choose between the two camps. The

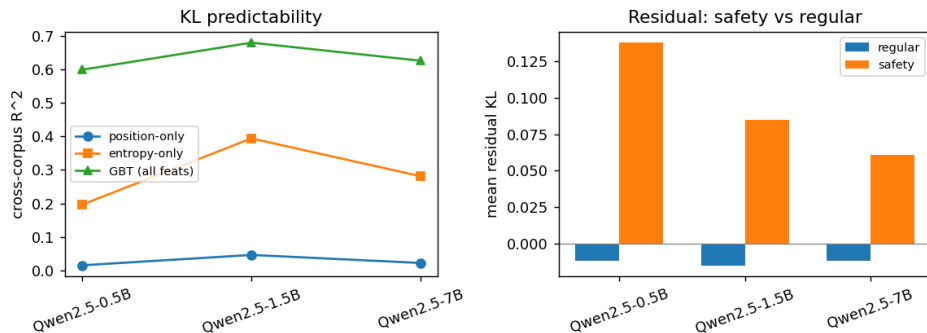


Figure 5: Summary across the three QWEN2.5 pairs (QWEN2.5-0.5B, QWEN2.5-1.5B, QWEN2.5-7B). Left to right: cross-corpus predictor R^2 (0.60–0.68), the safety-vs-regular residual gap (positive and significant in every pair, shrinking with scale), and the dominance of base entropy and base token surprise in feature importance. The bulk base-to-instruct shift is predictable from base-side context; what is left over is structured and alignment-specific.

predictable shift is superficial; the residual is where alignment writes something new, and our method separates them mechanically.

Safety integrates more smoothly as models grow. The safety residual is positive and significant at every scale, but its magnitude declines monotonically: rank-biserial $0.32 \rightarrow 0.30 \rightarrow 0.14$ from QWEN2.5-0.5B to QWEN2.5-1.5B to QWEN2.5-7B (table 3), with mean residual on safety tokens dropping from $+0.138$ to $+0.061$ while regular tokens stay near zero. The Mann–Whitney separation remains overwhelming throughout (p from ≈ 0 to 2×10^{-164}), so the effect is real at 7B—it is simply smaller. We read this as larger models integrating safety more smoothly into their base-conditioned predictions: the 7B base already places more probability mass on the cautious continuation, so flipping to a refusal or a hedge is less of a discontinuity and leaves a smaller surprise for the predictor. Safety in a bigger model is less of a bolt-on. The same trend appears in the per-string examples, where the 7B’s high-residual tokens (“Safety,” “guidelines,” “recommend,” “Instead”) are more explicitly policy-shaped than the 0.5B’s bluntly topical ones (“sex,” “violence,” “politics”), suggesting the small model still relies on coarse keyword triggers.

Limitations. We are scoring the instruct model’s own greedy decode, which is in-distribution for the instruct model and out-of-distribution for the base model—a confound Lin et al. [2023] flag for URIAL-style comparisons. We mitigate it two ways: we teacher-force the *identical* chat-formatted sequence through both models, so the only thing that differs is weights rather than context, and our analysis lives in the residual, which already factors out the base model’s own confidence on that sequence. A token that is merely unusual for the base model is well predicted by `base_tok_logprob` and therefore contributes little residual. That feature is informative by construction—the emitted token is the instruct model’s choice—but it is legitimate base-side structure, not leakage from the target: it is computed entirely from the base model’s distribution. Our token-class flags are regex- and vocabulary proxies, not a true part-of-speech tagger, so the class-level statistics (Kruskal–Wallis $p < 10^{-22}$ throughout) should be read as coarse groupings. All three pairs are Qwen2.5 instruct models trained with combined RLHF/DPO [Ouyang et al., 2022, Rafailov et al., 2023]; the gated SFT-only checkpoints we wanted were not downloadable, so we cannot separate SFT-driven from preference-optimization-driven divergence. Finally, we study greedy decoding capped at 80 new tokens. Longer or sampled generations could shift the position curve and surface late-sequence phenomena we do not see, and a single family at three sizes is a scale *trend*, not a scaling law.

The residual as an unsupervised alignment probe. Because the residual locates refusal, persona, and safety tokens without any labels, it is a candidate detector rather than only an analysis. The same teacher-forcing-plus-residual recipe extends directly to the logit-difference view of alignment used in proxy-tuning and Emulated Fine-Tuning, where the behavior delta is $\log P_{\text{inst}} - \log P_{\text{base}}$ [Liu et al., 2024, Mitchell et al., 2023]: our predictor models the expected magnitude of that delta from base-side context, so its residual flags positions where the delta is larger than the base model’s own uncertainty would explain. Practically, this gives an interpretability tool that highlights where an instruct model has been steered—which token in a generation is a refusal pivot, an identity assertion,

or a safety hedge—and an auditing signal for asking whether a deployed model’s safety behavior is smoothly integrated (small residual, as in the 7B) or brittle and keyword-triggered (large residual, as in the 0.5B). We see reading the residual as a general move: train a cheap predictor for the predictable part of a model difference, then study what it cannot explain.

6 Conclusion

Summary. We asked where instruction tuning changes a base model’s next-token distribution in ways a learned predictor cannot anticipate from simple base-side context, and what kinds of tokens those are. The answer is sharp. Per-token base→instruct divergence is mostly a predictable function of *base uncertainty*: a GBT trained only on base-side features recovers $R^2 \approx 0.60\text{--}0.68$ across all three QWEN2.5 pairs (Spearman ρ up to 0.89), with `base_tok_logprob` and `base_entropy` carrying nearly all the signal and position barely mattering. But the leftover residual is *not* noise. It is structured and alignment-specific: across all three scales it concentrates significantly on safety prompts (JUST-EVAL-INSTRUCT’s safety-tagged subset shows positive mean residual with rank-biserial up to 0.322, Mann–Whitney $p \rightarrow 0$) and on discourse/structural pivots and alignment-injected content—refusal-recovery turns (e.g., “If you have any other questions...”), persona and identity tokens (“Alibaba”), and knowledge-cutoff framing (“October”)—rather than the generic stylistic tokens that dominate the *predictable* part of the shift.

Takeaway. Most of what alignment does to the token distribution is anticipated by how uncertain the base model already was; what it does *beyond* that is a small, readable signature of safety, discourse control, and injected identity.

Future work. Several directions follow directly. (1) Score base-decoded or held-out reference sequences in addition to instruct-decoded ones, removing the which-model-generated-the-text confound from the residual. (2) Add an SFT-only model pair to separate the contributions of supervised fine-tuning from preference optimization. (3) Train a residual probe on base hidden states and attribute it with logit-lens tooling [Wang et al., 2025] to localize *where* in the network the unpredictable shift originates. (4) Use the residual itself as an unsupervised detector of alignment-driven behavior and validate it against held-out safety and persona labels.

References

- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. DoLa: Decoding by contrasting layers improves factuality in large language models. *arXiv preprint arXiv:2309.03883*, 2023.
- Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. Free dolly: Introducing the world’s first truly open instruction-tuned LLM. Databricks Blog, 2023.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization. *arXiv preprint arXiv:2210.15097*, 2022.
- Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base LLMs: Rethinking alignment via in-context learning. *arXiv preprint arXiv:2312.01552*, 2023.
- Alisa Liu, Xiaochuang Han, Yizhong Wang, Yulia Tsvetkov, Yejin Choi, and Noah A. Smith. Tuning language models by proxy. In *Conference on Language Modeling (COLM)*, 2024.
- Eric Mitchell, Rafael Rafailov, Archit Sharma, Chelsea Finn, and Christopher D. Manning. An emulator for fine-tuning large language models using small language models. *arXiv preprint arXiv:2310.12962*, 2023.

- Sean O’Brien and Mike Lewis. Contrastive decoding improves reasoning in large language models. *arXiv preprint arXiv:2309.09117*, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- Zhenyu Wang et al. LogitLens4LLMs: Extending logit lens analysis to modern large language models. *arXiv preprint arXiv:2503.11667*, 2025.
- Mohan Wu, Yousef Lubbad, et al. Revisiting the superficial alignment hypothesis. *arXiv preprint arXiv:2410.03717*, 2024.
- Taiqiang Wu et al. Shadow-FT: Tuning instruct models via training on paired base models. *arXiv preprint arXiv:2505.12716*, 2025.
- Andrew Yang, Belinda Z. Li, and Ari Holtzman. The geometry of probability concentration: Branching factor and the effect of alignment in large language models. *arXiv preprint arXiv:2506.17871*, 2026.
- Hao Zhao, Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. Is in-context learning sufficient for instruction following in LLMs? *arXiv preprint arXiv:2405.19874*, 2024.
- Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.